

ZiND!



Digital Africa AI 4 Startups Hackathon Programme

Final Report | March 2026



About Zindi

Zindi is the leading **data science and AI challenge platform** focused on Africa and emerging markets. Our platform brings together a **community of practice** in AI for Good, from researchers and students to entrepreneurs and professionals.

Zindi challenges feature **real data and problems** from partnerships with organisations around the world, and are hosted on our open and free platform. Users can **win prizes and gain real skills** while contributing their skills to challenges that matter. **Winning solutions** belong to the hosting organisation, and are often deployed post-challenge.

100k+
users

28%
women



About Zindi Challenges

A Zindi challenge is a **public competition**, with a dataset and a problem statement for anyone to try to solve. To solve a challenge, users make submissions and try to get the best score possible.

Scoring is achieved through an automated evaluation metric system, with **20+ scoring metrics** to choose from depending on the nature of the challenge. Users are ranked on a **live, dynamic leaderboard**.

Post-challenge, Zindi conducts a **final review** of solutions and KYC of winners to confirm their eligibility, and **solutions are transferred** to the partner organisations.

500+

challenges

\$1M+

prizes



Project Summary

Digital Africa's AI 4 Startups Hackathon Programme aims to facilitate access to AI models for African startups and the innovation ecosystem by organising five online challenges on Zindi, bringing together data scientists from across the continent.

The goal of these challenges is to produce AI models to answer the challenge identified in partnership with the startups. Developed models will be made open-source in accordance with grant funding conditions.

This project is part of the Data Governance in Africa Initiative, the African Union, and Expertise France.



Project Deliverables

In partnership with Digital Africa and startups across Africa, Zindi successfully delivered **15 open-source AI solutions** to real-world problems faced by **5 African startups**. These solutions were delivered through community-powered hackathons hosted on Zindi, which attracted **more than 4000 participants**.

To identify these challenges, Zindi and Digital Africa jointly put out several **calls for participation** to our respective communities and networks, and conducted **surveys of African startups**. From this pool of initial candidates, we conducted **25 data discovery workshops** with **18 startups** from Cote d'Ivoire, Ghana, Kenya, Malawi, South Africa, Senegal, and Tanzania.

Working with the 5 selected startups, we used their **data** to create **public custom challenges**, hosted on Zindi, and **awarded prizes** to the best solutions to each challenge.



Challenges overview

Challenge	Prizes	Startup	Topics
Côte d'Ivoire Byte-Sized Agriculture Challenge	€7000	Tolbi	earth observation, precision agriculture
agriBORA Commodity Price Forecasting Challenge	€8250	AgriBORA	price forecasting, agricultural economics
Farm to Feed Shopping Basket Recommendation Challenge	€8250	Farm to Feed	ecommerce, retail
DigiCow Farmer Training Adoption Challenge	€8250	DigiCow	extension, farmer training
Rhea Soil Nutrient Prediction Challenge	€8250	Rhea	soil health, predictive analytics

Challenge 1: Côte d'Ivoire Byte-Sized Agriculture Challenge

Can you use satellite data to classify pixels for crop identification in West Africa?

Startup: [Tolbi](#)

Country: Côte d'Ivoire

Prizes: €7000

Evaluation: F1 (Accuracy)

[Challenge page on Zindi](#)

Tolbi's mission is to advance precision agriculture and nature restoration across Africa. The challenge aimed to develop a pixel-level classification model that can distinguish between cocoa, rubber, and oil palm plantations using open-source earth observation data.

This solution directly supports Tolbi's efforts to scale precision agriculture tools across Côte d'Ivoire and beyond, enabling smarter land use, better resource management, and more sustainable agricultural practices across West Africa.

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The poster features a blue background with a green pattern of small white triangles on the left. At the top, it displays the "Global Gateway" logo (a circle of stars) and the "European Union" and "African Union" logos. The main text reads "PRIZE POOL | € 7000" and "Côte d'Ivoire Byte-Sized Agriculture Challenge". Below this, it asks "Can you use satellite data to classify pixels for crop identification in West Africa?". A horizontal strip of satellite imagery shows various agricultural landscapes. At the bottom left, a green box contains the text "DATA GOVERNANCE IN AFRICA". The bottom right corner features logos for "D4D HUB", "Digital Africa", "Tolbi", and "ZiND!".

Global Gateway

European Union African Union

PRIZE POOL | € 7000

Côte d'Ivoire Byte-Sized Agriculture Challenge

Can you use satellite data to classify pixels for crop identification in West Africa?

DATA GOVERNANCE IN AFRICA

D4D HUB

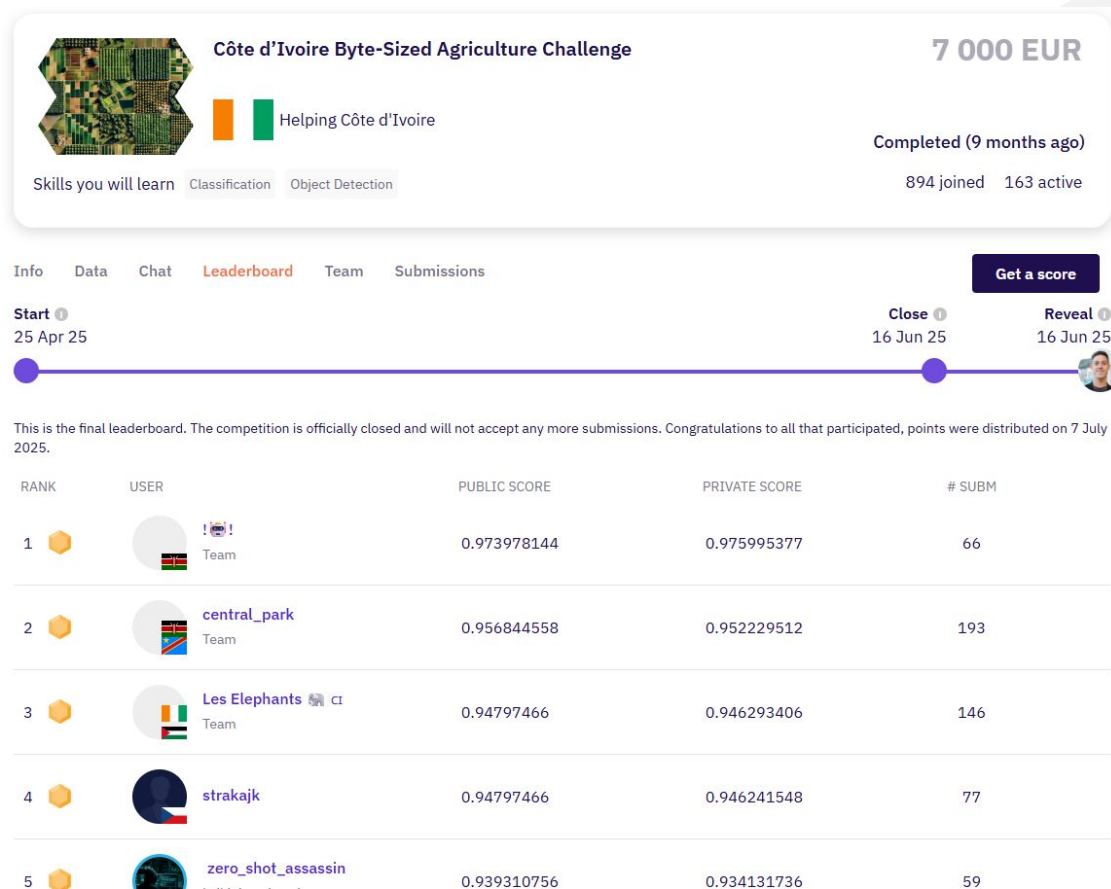
Digital Africa

Tolbi

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Challenge 1: Statistics

- **866** users engaged
- **16%** female
- **36** African countries (**27** French-speaking)
- **6 463** submissions
- **18 000+** estimated hours of data science work supported
- High enrolment from **Côte d'Ivoire**



Challenge 1: Winners



Top overall model: Darius Moruri (**Brainiac**)

Prize: €3500

Country: Kenya

Medals: 24th Gold

F1-score: 0.97599



Top-placed female: Jane Njuguna (**Jeshen**)

Prize: €2000

Country: Kenya

Medals: 1st Gold

F1-score: 0.85734



Top Ivorian: Mohammed Tuo (**Les Elephants**)

Prize: €1500

Country: Ivory Coast & Tunisia

Medals: 32nd & 55th Gold

F1-score: 0.94629

Challenge 1: Solutions



Top overall model: Darius Moruri (**Brainiac**)

This solution combines ConvNeXt deep learning architecture with LightGBM ensemble methods.



Top-placed female: Jane Njuguna (**Jeshen**)

This solution uses a Light Gradient Boosting Machine (LightGBM) model for predicting crop type.



Top Ivorian: Mohomed Tuo (**Les Elephants**)

This solution relies on an ensemble of transformer-based architectures to classify crop types using monthly satellite images from Sentinel-2.

Your winning solutions

Challenge 2: agriBORA Commodity Price Forecasting Challenge

Can you accurately forecast weekly market prices for maize in Kenya?

Startup: [agriBORA](#)

Country: Kenya

Prizes: €8250

Evaluation: Mean Absolute Error (MAE) + Root Mean Square Error (RMSE)

[Challenge page on Zindi](#)

agriBORA helps smallholder farmers in Kenya by offering certified warehouses where farmers can safely store their produce and receive digital warehouse certificates, enabling access to loans and the option of delayed selling. Using historical prices of dry maize in Kenya, your task is to develop a machine learning solution to predict average weekly prices of maize Kenya.

Accurate forecasts will help farmers time their sales effectively, increase earnings, and strengthen agriBORA's services to East African farmers.

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PRIZE POOL | \$ 8250

agriBORA Commodity Price Forecasting Challenge

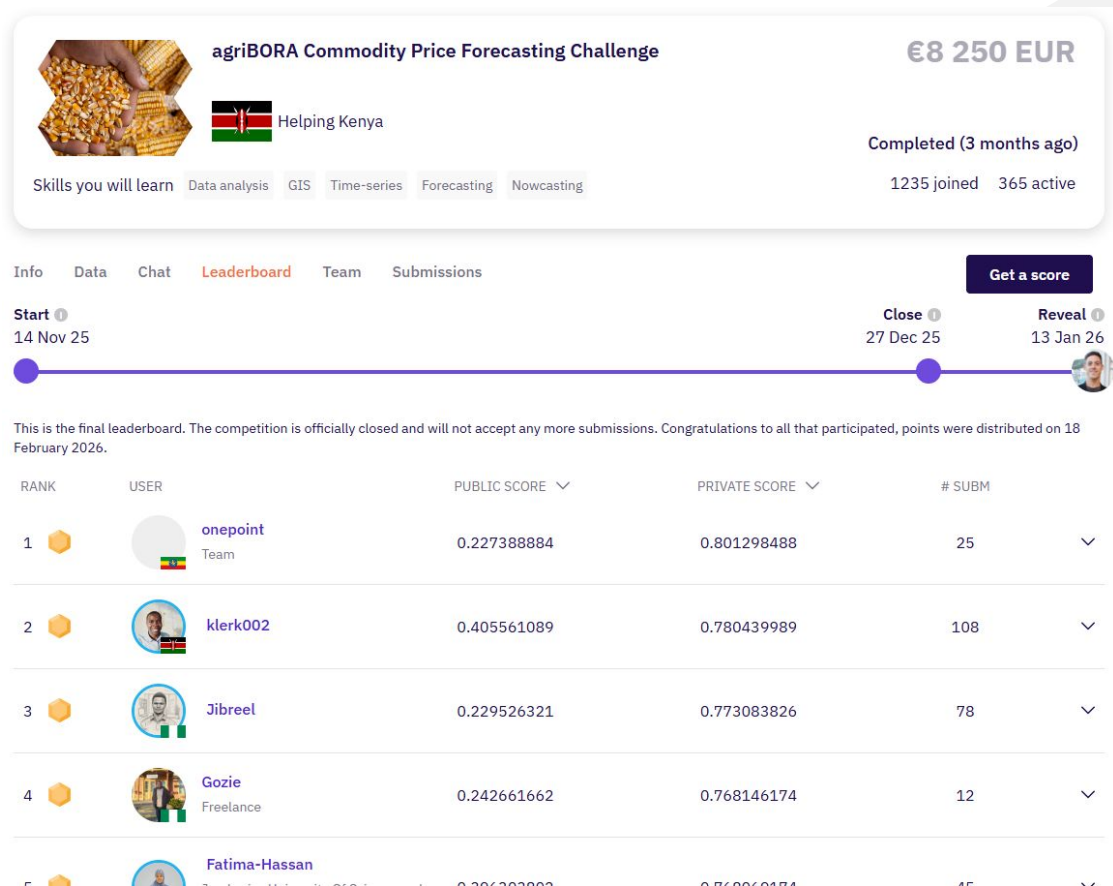
Can you accurately forecast weekly market prices for maize in Kenya?

DATA
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IN AFRICA



Challenge 2: Statistics

- **1095** users engaged
- **14.5%** female
- **78** countries (**36** African)
- **4 421** submissions
- **13 000+** estimated hours of data science work supported
- Winning solutions represented **diverse approaches** to the same problem



Challenge 2: Winners



Top overall model: Yisak Bule ([Yisakberhanu](#))

Prize: €4000

Country: Ethiopia

Medals: 12th Gold

Multi-score: 0.800



Top-placed female: Fatima Hassan Mohamed ([Fatima-Hassan](#))

Prize: €2750

Country: Somalia

Medals: 1st Gold

Multi-score: 0.768



Top Kenyan: Chrispinus Owino ([klerk002](#))

Prize: €1500

Country: Kenya

Medals: 1st Gold

Multi-score: 0.780

Challenge 2: Solutions



Top overall model: Yisak Bule ([Yisakberhanu](#))

Forecast weekly price changes (momentum) rather than absolute prices, using recent market movement as the strongest predictor.



Top-placed female: Fatima Hassan Mohamed ([Fatima-Hassan](#))

Integrated historical agriBORA data with recent updates, and integrated external KAMIS market indicators (wholesale, retail, supply trends).



Top Kenyan: Chrispinus Owino ([klerk002](#))

Cascading prediction architecture models how maize prices move through the supply chain.

[Your winning solutions](#)

Challenge 3: Farm to Feed Shopping Basket Recommendation Challenge

Can you predict future shopping baskets to recommend better fresh produce purchases?

Startup: [Farm to Feed](#)

Country: Kenya

Prizes: €8250

Evaluation: Area Under the Curve (AUC) and Mean Absolute Error (MAE)

[Challenge page on Zindi](#)

Farm to Feed connects farmers, businesses and consumers with odd-looking fruits and vegetables that would otherwise go to waste. This challenge aims to build a recommender system for their ecommerce app using historical transaction data.

The winning models will help Farm to Feed move surplus and underused produce efficiently, reduce food waste, and expand market access for smallholder farmers, while improving operations, marketing, and sales for the business.

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PRIZE POOL | \$ 8250

Farm to Feed Shopping Basket Recommendation Challenge

Can you predict future shopping baskets to recommend better fresh produce purchases?

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Challenge 3: Statistics

- **733** users engaged
- **18%** female
- **68** countries (**36** African)
- **5108** submissions
- **15 000+** estimated hours of data science work supported
- High enrolment from **Kenya**



Farm to Feed Shopping Basket Recommendation Challenge

Helping Kenya

Skills you will learn: Machine Learning, Prediction, Feature Engineering

€8 250 EUR

Completed (2 months ago)

746 joined 266 active

Info Data Chat **Leaderboard** [Accept terms](#)

Start 02 Dec 25 **Close** 19 Jan 26 **Reveal** 20 Jan 26

This is the final leaderboard. The competition is officially closed and will not accept any more submissions. Congratulations to all that participated, points were distributed on 18 February 2026.

RANK	USER	PUBLIC SCORE	PRIVATE SCORE	# SUBM
1	 learnagain Team	0.980763092	0.973387989	66
2	 shamba_shapeup Team	0.982296581	0.972750766	63
3	 Trial Team	0.980759301	0.972050096	207
4	 israelkingz University of ibadan	0.979937735	0.971567361	7
5	 Bomb Squad	0.980885882	0.971529949	125

Challenge 3: Winners



Top overall model: Yisak Bule ([Yisakberhanu](#))

Prize: €4000

Country: Ethiopia

Medals: 12th Gold

Multi-score: 0.800



Top-placed female team: Laurah Agosa, Satelite Alison Ndayikunda, Mourine Wamuyu ([LiveFeed](#))

Prize: €2750

Country: Kenya

Medals: 1st, 2nd, 3rd Gold

Multi-score: 0.9701



Top Kenyan: Steven Kolesh ([shamba_shapeup](#))

Prize: €1500

Country: Kenya, Ivory Coast

Medals: 47th, 20th Gold

Multi-score: 0.9727

Challenge 3: Solutions



Top overall model: Yisak Bule ([Yisakberhanu](#))

Combines machine learning and statistical heuristics with multi-seed ensembling and Test-Time Augmentation (TTA).



Top-placed female team: Laura Agosa, Satelite Alison Ndayikunda, Mourine Wamuyu ([LiveFeed](#))

Multi-target forecasting: Predicts purchase likelihood (7 & 14 days) and expected quantities using a dual classification-regression setup.



Top Kenyan: Steven Kolesh ([shamba_shapeup](#))

Dual gradient boosting models (LightGBM + CatBoost GPU) combined with deep temporal feature engineering. Production-ready, reproducible pipeline.

Your winning solutions

Challenge 4: DigiCow Farmer Training Adoption Challenge

Can you predict which farmers will turn training into action?

Startup: [DigiCow](#)

Country: Kenya

Prizes: €8250

Evaluation: Log Loss and ROC-AUC

[Challenge page on Zindi](#)

DigiCow supports smallholder farmers through training programmes. Early adoption prediction enables prioritised follow-up for at-risk farmers, tailored support based on likelihood of action, more effective extension programme design, and better resource allocation for maximum impact.

The challenge objective was to predict the probabilities that a farmer will adopt a practice within 7, 90, and 120 days after training, using only information available at the time of training.



The poster features a blue background with a green patterned vertical bar on the left. At the top, it displays the 'Global Gateway' logo and the European Union and African Union flags. The main text reads 'PRIZE POOL | € 8 250' and 'DigiCow Farmer Training Adoption Challenge'. Below this, it asks 'Can you predict which farmers will turn training into action?'. A photograph of a smiling farmer with a cow is at the bottom. A green box on the left side of the photo contains the text 'DATA GOVERNANCE IN AFRICA'. The footer includes logos for D4D HUB, Digital Africa, EXPERTISE FRANCE GROUPE AFD, DigiCow Africa, and ZiND!

Global Gateway

PRIZE POOL | € 8 250

DigiCow Farmer Training Adoption Challenge

Can you predict which farmers will turn training into action?

DATA GOVERNANCE IN AFRICA

D4D HUB

Digital Africa

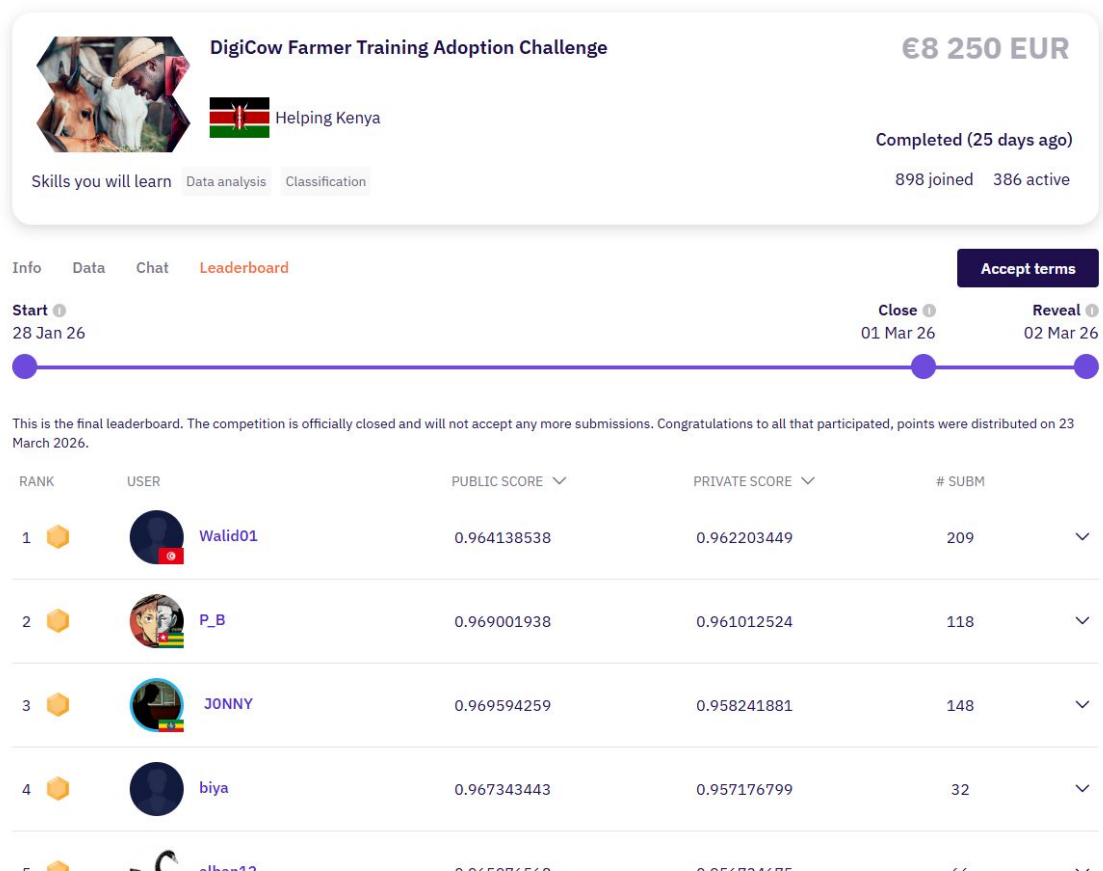
EXPERTISE FRANCE GROUPE AFD

DigiCow Africa

ZiND!

Challenge 4: Statistics

- **896** users engaged
- **16%** female
- **83** countries (**36** African)
- **20 050** submissions
- **60 000+** estimated hours of data science work supported
- All solutions relied on **ensembling** to achieve their high scores.



Challenge 4: Winners



Top overall model:
Missodey Raoul (P_B)

Prize: €4000

Country: Togo

Medals: 3rd Gold

Multi-score: 0.9610



Top-placed female: Abinet
Bekele (biya)

Prize: €2750

Country: Ethiopia

Medals: 2nd Gold

Multi-score: 0.9571



Top Kenyan: Moan Lobago
(CN-Boost)

Prize: €1500

Country: Kenya, Nigeria

Medals: 8th & 8th Gold

Multi-score: 0.9537

Challenge 4: Winners



Top overall model:
Missodey Raoul (P_B)

Multi-model ensemble optimised separately for each evaluation metric.



Top-placed female: Abinet Bekele (biya)

Two-model ensemble (XGBoost + LightGBM) optimised separately for each metric.



Top Kenyan: Moan Lobago (CN-Boost)

Four-model voting ensemble (XGBoost, LightGBM, CatBoost, and HistGradientBoosting) with a single prediction output.

Your winning solutions

Challenge 5: Rhea Soil Nutrient Prediction Challenge

Can you predict soil nutrient content when lab tests aren't an option?

Startup: [Rhea](#)

Country: Kenya

Prizes: €8250

Evaluation: Root Mean Squared Error (RMSE)

[Challenge page on Zindi](#)

Soil health is fundamental to crop yields, fertiliser efficiency, climate resilience and long-term sustainability of farming systems. However, reliable soil testing remains severely limited across Africa, and traditional laboratory analysis is expensive, slow, and often unavailable to smallholder farmers. This challenge aims to help bridge the gap using soil data and machine learning.

The winning models will help Rhea assess soil status, helping farmers make informed decisions with tailored recommendations, ensuring sustainable and profitable farming.

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PRIZE POOL | € 8 250

Rhea Soil Nutrient Prediction Challenge

Can you predict soil nutrient content when lab tests aren't an option?

DATA
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IN AFRICA



Challenge 5: Statistics

- › **610** users engaged
- › **18%** female
- › **82** countries (**38** African countries)
- › **4 441** submissions
- › **15 000+** estimated hours of data science work supported
- › **Highest female participation rate** of all challenges



Rhea Soil Nutrient Prediction Challenge

Helping Africa

Skills you will learn [GIS](#) [Visualisation](#) [Prediction](#)

€8 250 EUR

Under code review

603 joined 237 active

[Info](#) [Data](#) [Leaderboard](#) [Accept terms](#)

Start ⓘ
04 Feb 26

Close ⓘ
08 Mar 26

Reveal ⓘ
09 Mar 26

This leaderboard was posted upon the close of the competition and reflects the results of the private leaderboard (the public leaderboard which was open for the duration of the competition reflected only a portion of the test data). Submissions are no longer accepted. Scores and ranking are subject to change until 30 March 2026 pending final review of winning solutions.

RANK	USER	PUBLIC SCORE ▾	PRIVATE SCORE ▾	# SUBM	
1	 ML_vibe Team	0.631267328	0.55562599	67	▾
2	 rynstensis	0.525595519	0.500202915	24	▾
3	 lamino14	0.470535046	0.484457813	14	▾
4	 BitsnBytes Team	0.625591982	0.483902223	118	▾
5	 keystats Mount Kenya University	0.451637314	0.472775654	72	▾

Challenge 5: Winners



Top overall model: Yisak Bule ([Yisakberhanu](#))

Prize: €4000

Country: Ethiopia

Medals: 17th Gold

Multi-score: 0.5556



Top-placed female: Gladys Losute ([BitsnBytes](#))

Prize: €2750

Country: Kenya

Medals: 3rd Gold

Multi-score: 0.4839



Top Kenyan: Jackson Kahungu ([keystats](#))

Prize: €1500

Country: Kenya

Medals: 7th Gold

Multi-score: 0.4727

Challenge 5: Winners



Top overall model: Yisak Bule ([Yisakberhanu](#))

Domain-aware ensemble of Random Forest + LightGBM, using adversarial feature selection and a separate pipeline for historical and 2025 data during inference.



Top-placed female: Gladys Losute ([BitsnBytes](#))

Used climate-based clustering (Köppen–Geiger zones) and village-level modelling with hierarchical statistics with an ensemble of diverse models.



Top Kenyan: Jackson Kahungu ([keystats](#))

Combined excellent domain knowledge with 1,000+ features from satellite, climate, and soil data and soil-depth matching for a unique approach.

[Your winning solutions](#)

Lessons Learned

In the African startup space, data maturity matters: When interviewing startups, it very quickly became clear that we were dealing with a very wide array of data maturity, from teams with fully fleshed out data functions, to startups who had all their data in a single spreadsheet and just wanted to use AI for something without understanding it. This required a differentiated approach, depending on our assessment of the startup's level at the beginning of the process.

Market maturity differs hugely across Africa: South Africa and Kenya showed by far the most data maturity in the markets we operated in. Beyond those two countries, most startups had little to no data expertise or readiness for an AI solution. This presented a challenge to the programme, and highlights that the hackathon format only works with a certain level of organisation. The challenges that worked best were with agriBORA, Farm to Feed, and Rhea, all of which have an advanced understanding of the value of their data, and the enthusiasm and expertise to get valuable solutions from their challenge.

Open-source was a blocker for several startups: Many startups pulled out of the process after initial enthusiasm, when it became clear that they would have to share some data to participate, and that the final solutions would be open-sourced. In some cases, this reflects a misguided concern (often from funders or the startups board) that the start-up would lose competitive advantage by having these solutions open-sourced. In other cases, the concern was around data privacy, which is a valid concern that many organisations face when participating in an AI hackathon in Zindi.

Lessons Learned

Workshop methodology improved over time: Initial workshops with startups were complex and didn't go smoothly. We improved our methodology over time to get the most out of a short time with a startup. This included: better preparation based on the answers provided in the survey, taking a product development approach to explore opportunities and challenges, and taking more time to explain data requirements for a challenge of this nature. We also took additional care to explain the open-source requirements of the programme up-front, to make sure there were no concerns around data privacy

Dedicated prizes worked: Setting aside dedicated prize money for top performances from women and at a country level drove consistently higher participation from these groups in all challenges. This was a great result for the project, as these wins empower a broader group of people than would normally be the case. This empowerment is three-fold: firstly, winning money is a direct benefit. Secondly, the win improves confidence and self-efficacy in the winner, leading to compounding positive effects in their career. Lastly, organisers often want to set up calls with the winners, and sometimes contract them for further work implementing the solution. This provides the with work experience and further career development.



Thank you!

